

Climate change adaptation: Challenges for agricultural sustainability

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Funding information

Science and Technology Major Project of Guangxi, Grant/Award Number: Guike AA22117002-1; Fund of Guangxi Academy of Agricultural Sciences, Grant/Award Number: 2021YT011; Guangxi Characteristic Crop Experimental Station, Grant/Award Number: GTS2022022; Guangxi Innovation Teams of Modern Agriculture Technology, Grant/Award Number: nycytxgxcxtd-2021-03; National Natural Science Foundation of China, Grant/Award Number: 31760415; National Key Research and Development Project, Grant/Award Number: 2022YFD2301102-07; Guangxi Natural Science Foundation, Grant/Award Number: 2021GXNSFAA220022

Abstract

Climate change poses a substantial threat to agricultural sustainability globally. Agriculture is a vital component of the gross domestic production of developing countries. The multifaceted impacts of climate change on agriculture, highlighting how extreme weather events such as water stress, heatwaves, erratic rainfall, storms, floods, and emerging pest infestations are disrupting agricultural productivity. The socioeconomic status of farmers is particularly vulnerable to climatic extremes with future projections indicating significant increment in ambient air temperatures and unpredictable, intense rainfall patterns. Agriculture has historically relied on the extensive use of synthetic fertilizers, herbicides, and insecticides, combined with advancements in irrigation and biotechnological approaches to boost productivity. It encompasses a range of practices designed to enhance the resilience of agricultural systems, improve productivity, and reduce greenhouse gas emissions. By adopting climate-smart practices, farmers can better adapt to changing climatic conditions, thereby ensuring more sustainable and secure food production. Furthermore, it identifies key areas for future research, focusing on the development of innovative adaptation and mitigation strategies. These strategies are essential for minimizing the detrimental impacts of climate change on agriculture and for promoting the long-term sustainability of food systems. This article underscores the importance of interdisciplinary approaches and the integration of advanced technologies to address the challenges posed by climate change. By fostering a deeper understanding of these issues to inform policymakers, researchers, and practitioners

about effective strategies to safeguard agricultural productivity and food security in the face of changing climate.

KEYWORDS

adaptation, food security, impact of climate change, nutrition, restoration strategy

1 | INTRODUCTION

Adverse climate conditions negatively influence agricultural crop productivity, human health, and agroecosystems and food security (Altieri & Nicholls, 2017; Kumari et al., 2022, 2023; Raza et al., 2019; Verma, Song, et al., 2020; Verma, Singh, et al., 2020). Further, fast urbanization and population enhancement may culminate into severity to make available food and feed, adequately, as FAO estimate suggested that food production may increase by nearly 60% in the upcoming years (Altieri & Nicholls, 2017; FAO, 2023; Verma, Song, Verma, et al., 2023). The United Nations (2015) estimated that 836 million people worldwide were still in extreme poverty in 2015. As of 2022, the Food and Agriculture Organization (FAO) estimated that between 657 million and 676 million people were living in extreme poverty, with slowly decreasing poverty at a rate of 2% annually (FAO, 2022).

Worldwide, people living in rural areas are more exposed to food insecurity owing to limited access to food and financial resources. Nearly 50% are small farmers, producing on marginal lands that are particularly sensitive to the negative effects of climatic variables, that is, water stress and soil flooding conditions. Nearly 2 billion people are supported by 500 million smallholder farms in developing countries. In Asia and sub-Saharan Africa, these small farms provide almost 80% of the food consumed (IFAD, 2011; Khan et al., 2023; Verma, Song, Degu, et al., 2023). Forests are a source of income for rural areas (Arbuckle JG et al., 2015; World Bank, 2002). Between 660 and 820 million people depend entirely or partly on fisheries, aquaculture, and related businesses for their livelihood and sustenance (Reckling et al., 2018) with food security, that is, “everyone having physical and economic access to enough safe food for an active and better life” (Brown & Berry, 2022; FAO, 2023; Otto et al., 2017).

Ecosystems, plants, and animals have adapted to local climatic variables, globally. When these conditions become more unfavourable, then, native flora and fauna can be affected by reducing productivity or even losing the productivity (Gray & Brady, 2016; Jagadesh et al., 2024; Kumari et al., 2023; Lu et al., 2015; Verma, Song, Tian, et al., 2021). The direct impact of climatic conditions on entire agroecosystems is challenging to forecast because each component will respond differently and associate with the others. Under controlled conditions, various plants respond favourably to increased atmospheric CO₂ levels (Figure 1). But in uncontrolled conditions crop loss leads to farmers' distress, inflation, and far-reaching economic consequences (Kumari et al., 2022; Lusk & Polzin, 2023; Patni et al., 2022). Hence, the main objectives of this review are to assess the impacts of climate change on sustainable agriculture and to

discuss opportunities for adaptation and mitigation approaches for reducing the drastic effects of climate change.

2 | INFLUENCE OF CLIMATE CHANGE ON PLANT AGROECOSYSTEMS

Climate change impacts agricultural production systems in both direct and indirect ways. Variations in physical activities, such as ambient air temperature, distributions throughout the year and availability of water, directly affect agricultural productivity (Abbas et al., 2022; Al-Tawaha et al., 2020; Ray et al., 2019; Shrestha, 2019; Zhao et al., 2017; Verma, Song, Lin, et al., 2021; Verma, Song, Zeng, et al., 2021). Pale climatic trends have been observed to impact agricultural productivity at different locations, with harmful effects in comparison to positive effects (Bajwa et al., 2020; Khan et al., 2023; Ray et al., 2015). Wheat and maize crops have already been adversely affected by both global and regional climate change in terms of yield and quality (Espeland & Kettenring, 2018; Kumari et al., 2014). There are implications for rice quality and yield due to the increased frequency of sweltering nights in most areas. Using different approaches and a wide range of crop models, it is feasible to estimate how upcoming climate change may affect agricultural crop yield (Table 1) (FAO, 2017; Frieler et al., 2015; Gray & Brady, 2016; Kumari et al., 2022, 2023; Verma, Song, Yadav, et al., 2022; Verma, Song, Budeguer, et al., 2022), as climate change is anticipated to reduce the productivity of crops, such as wheat, rice, and maize crops, in low-latitude and desert areas (Abbass et al., 2022; Bajwa et al., 2020; FAO, 2023).

The effects of climate change on mid- to high-latitude regions are anticipated to be more variable under higher warming conditions (IPCC, 2014). Limited high-latitude locations may benefit significantly from warmer temperatures and long-duration growing seasons, but other environmental concerns include low soil fertility (Brown & Berry, 2022; FAO, 2015, 2023; Hussain et al., 2020; Meierrieks, 2021; Ray et al., 2015). Regional and subregional spatial inequalities are also observed in places with considerable differences in elevation, and the effects between high- and low-latitude areas show that imbalances between developed and developing nations will likely worsen due to climate change (Elbehri et al., 2015). Crop yield variability will increase across diverse locations due to climate change. For example, in major breadbasket regions, climate variability can explain over 60% of yield variability. This suggests that as climate change progresses, we can expect increased variability in crop yields due to more extreme weather events, changing precipitation patterns, and temperature fluctuations

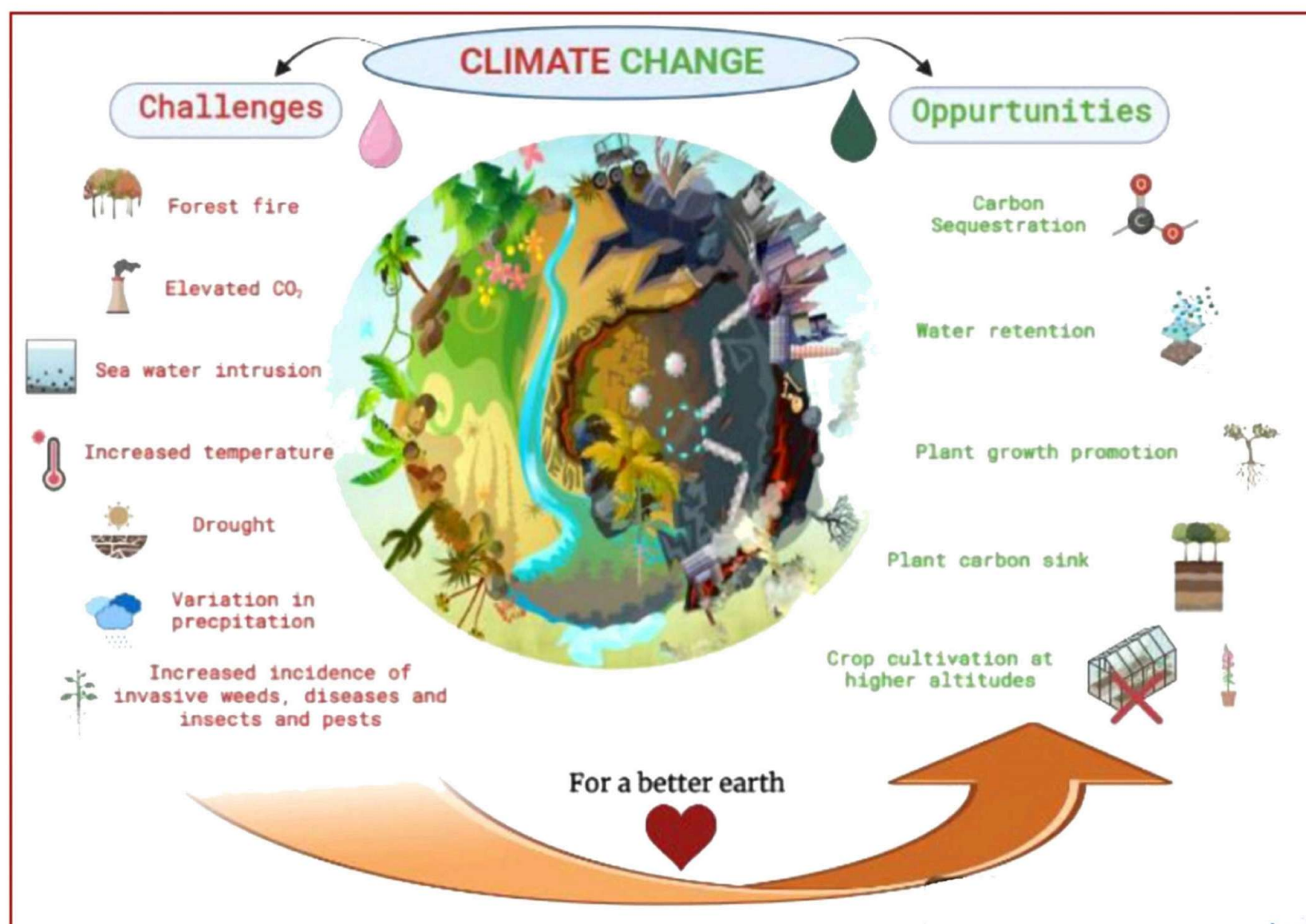


FIGURE 1 Challenges and opportunities associated with climate change for sustainable agricultural production.

(Challinor et al., 2014). The effect of temperature on proper plant establishment found to be negatively correlated (Kumari et al., 2022; Malhi et al., 2021).

The stimulatory effect of elevated atmospheric CO₂ on plant development (enhancing photosynthetic responses and improving water-use efficiency) is being observed in most cases. The impact of climate change on crop development depends on various climatic factors, such as temperature, precipitation patterns, and atmospheric CO₂ concentrations (Datta et al., 2022; Seth, 2014; Seth et al., 2014; Zhao et al., 2017; Verma, Song, Li, et al., 2022; Verma, Song, Joshi, Rajput, et al., 2022; Verma, Song, Joshi, Tian, et al., 2022). The recent studies have demonstrated that high tropospheric ozone concentrations have a detrimental effect on crop productivity, as estimated by crop production damage in *Glycine max* (8.5%–14%), *Triticum* spp. (4%–15%), and *Zea mays* (2.2%–5.5%) (Lusk & Polzin, 2023; Ray et al., 2015). Currently, the impact of global climatic variations on agri-crops is assessed within the framework of the Agricultural Model Intercomparison and Improvement Project (AgMIP) and the Inter-Sectoral Impact Model. Plant productivity varies between –20% and –45% (*Zea mays*), –5% and –50% (*Triticum* spp.), –20% and –30% (*Oryza sativa*), and –30% and –60% (*Glycine max*) (Martinez

et al., 2018; Muller & Elliott, 2015), may also be rated positive (Thomas & Rosegrant, 2015).

The studies based on climate and crop models suggest that the agroecological potential of the grain-producing zone of Central Eurasia may increase due to warm temperatures, long growing seasons and favourable effects of high atmospheric CO₂ concentrations over the crops while, the loss is predicted in agricultural efficiency during frequent droughts (Challinor et al., 2018; Lioubimtseva et al., 2015; Malhi et al., 2021; Zhao et al., 2017; Verma, Singh, et al., 2020; Verma, Song, Verma, et al., 2023; Verma, Song, Degu, et al., 2023). However, the agri-zones having increase in winter temperatures may prolong the growing season with improved CO₂ fertilization (Becklin et al., 2016; Habib-ur-Rahman et al., 2022; Lesk et al., 2016; Zhao et al., 2017). Recent research findings suggest that cassava plants have an ideal high temperature for photosynthetic leaf gas exchange and proper plant development and a favourable response to rising CO₂ levels (Kumari et al., 2022), apple and cherry plants have adverse effects during climate change (Salama et al., 2021), the production of grapevine may decline (Khan et al., 2023; Ray et al., 2015). Figure 2 depicts the influence of climate change on different ecosystems.

TABLE 1 Outline of several of the tools available for climate change adaptation, categorized by their primary functions.

Category	Application	Impact
Climate Modelling	Climate Models	Simulate climate scenarios to assess future trends.
	Downscaling Models	Refine global climate data for regional predictions.
Climate Data	Climate Databases	Provide historical and projected climate data.
	Climate Information Systems	Store and distribute climate data for analysis.
Risk Assessment	Vulnerability Assessments	Identify areas and systems at risk from climate change.
	Impact Assessments	Evaluate potential consequences on ecosystems.
Ecosystem Assessment	Risk Mapping Tools	Create maps displaying climate-related risks.
	Ecosystem Services Assessment	Assess the value of ecosystem services for adaptation.
	Biodiversity Monitoring	Track changes in species and habitat distributions.
Soil Health Assessment	Habitat Suitability Models	Predict suitable areas for species and ecosystems.
	Soil Quality Assessment	Evaluate soil properties and health indicators.
	Soil Erosion Models	Predict erosion risk and soil loss.
Water Management	Hydrological Models	Simulate water flow and availability.
	Irrigation Decision Support	Assist in optimizing irrigation practices.
Carbon Sequestration	Carbon Accounting Tools	Estimate carbon storage and emissions.
	Land Use Change Models	Predict carbon impacts of land management changes.
Adaptation Planning	Climate Adaptation Plans	Develop strategies for adapting to climate change.
	Agroecological Planning	Design farming systems for climate resilience.
Policy and Decision Support	Policy Evaluation Tools	Analyse the effectiveness of climate policies.
	Decision Support Systems (DSS)	Provide information for informed decision-making.
Communication	GIS and Mapping Software	Create maps to visualize climate data and impacts.
	Stakeholder Engagement Platforms	Facilitate communication and collaboration.

Different forms of information, such as expert knowledge and experimental investigations are used to estimate the consequences of climate change on plant development (Kumar & Seth, 2022; Marcinkowski & Piniewski, 2018). Variations in temperature and CO₂ levels may cause agronomic crops and invasive weeds to become more widely distributed and competitive. Crop models can potentially describe the internal processes of crops by considering temperature increases and elevated CO₂ during critical growth stages of various crops (Challinor et al., 2018; Karavolias et al., 2021; Zhao et al., 2017).

Pests are more significantly impacted by climate change since suitable warm temperatures help them to establish themselves. Climate change may have a more detrimental effect on humans by causing pests to emerge earlier in the season due to elevated temperatures. The most destructive migratory pests and desert locusts are projected to experience significant alterations due to changes in temperature, rainfall, and wind patterns (Cressman, 2013; Datta et al., 2022; Jupinder et al., 2014; Verma, Song, Zeng, et al., 2021; Verma, Song, Yadav, et al., 2022; Verma, Song, Joshi, Tian, et al., 2022). In India, major areas of Rajasthan have already experienced an outbreak of desert locusts and a devastating impact on crops, also found in nine other states, namely, Madhya Pradesh,

Punjab, Gujarat, Uttar Pradesh, Maharashtra, Chhattisgarh, Bihar, Haryana, and Uttarakhand, also reported locust incursions. In these states, efforts toward locust management were carried out in collaboration with state governments.

The significant diseases wheat stem and stripe rusts, and the three main meteorological variables, moisture, temperature, and wind, can cause epidemics (Luck et al., 2011). It causes severe outbreaks and major damage to wheat production, and these outbreaks are most severe in East Central Asia and Eastern and Northern Africa. Additionally, the international trade of millions of plants and plant-based products, including grains, vegetables, fruits, and wood, facilitates the migration of plant pests. The risk of pest contamination increases as more diverse commodities are transported to an increasing number of nations (As per FAO, International Standards for Phytosanitary Measures and IPPC, International Plant Protection Convention). The increased vulnerability of plants to the direct consequences of climatic changes has led to damage and severe adverse effects due to the classic triangle between plant hosts, pathogens, and the environment (Bailey et al., 2015; Brown & Berry, 2022; Jagadesh et al., 2023, 2024; Kumari et al., 2023; Marcinkowski & Piniewski, 2018).

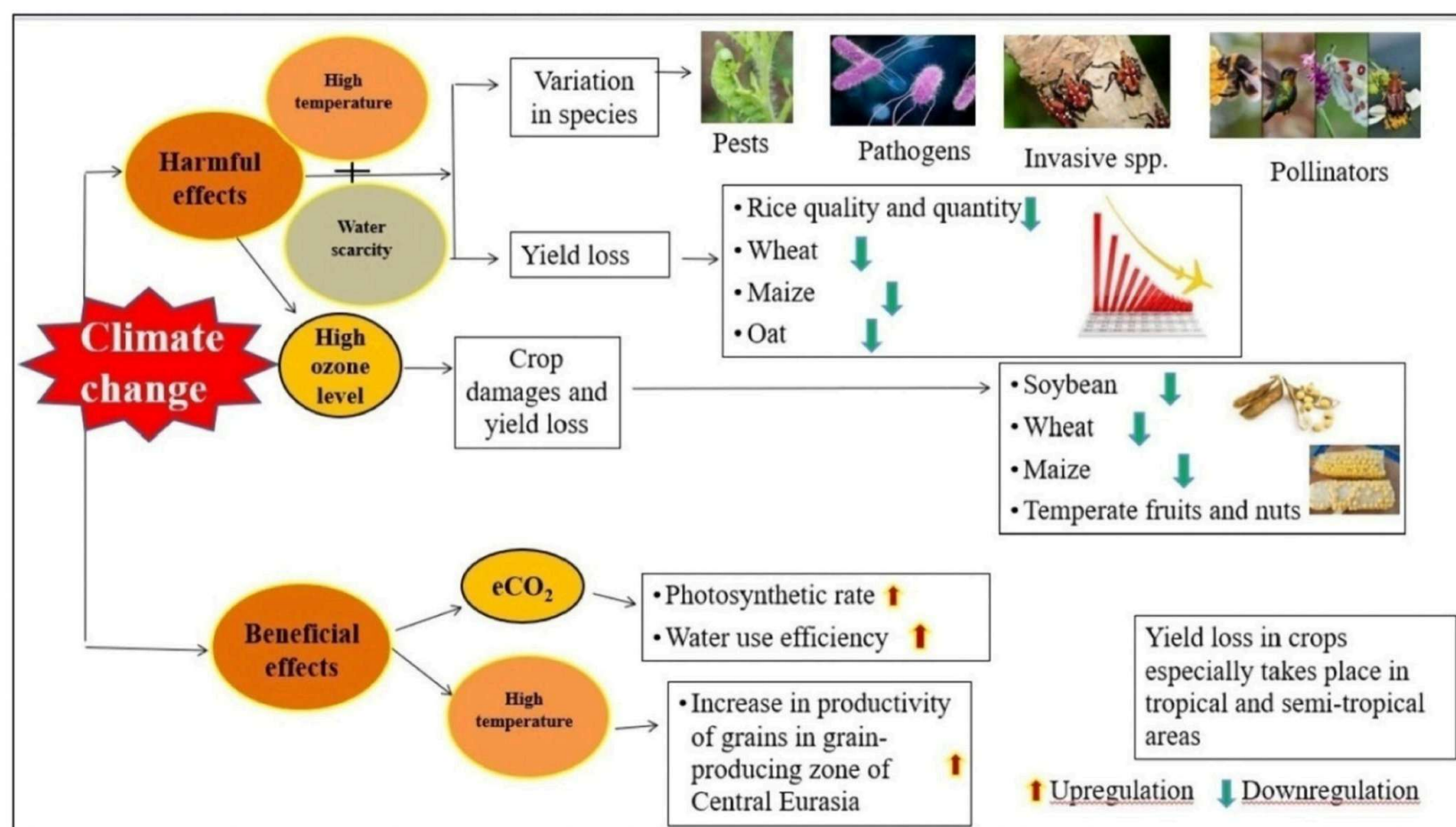


FIGURE 2 Influence of climate change on various agroecosystems. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pce.15078)]

The range and characteristics of pollinators and other species are crucial to production because of their ecological services (Bahinipati et al., 2021; Khan et al., 2023). Thirty-five percent of global agricultural productivity is impacted by pollinators such as bees, birds, and bats, raising yields of 87 of the top food crops and medicinal plants (Dhanya & Ramachandran, 2016). The symbiosis between pollinators and fertilized species, as well as other factors, plays a significant role in pollination. It frequently results from complex interactions between plants and animals, and a decline or extinction of either will have an impact on the survival of the others. It is a growing concern that the effects of climate change could impair this relationship. Most pollinators are already inside their ideal temperature tolerance range in the tropics. The physiological activities of plants and insect pollinators and the effects of warming on phenological and distributional modifications have all been extensively studied (Abbass et al., 2022; Cressman, 2013; Jupinder et al., 2014; Verma, Song, Budeguer, et al., 2022; Verma, Song, Li, et al., 2022; Verma, Song, Joshi, Rajput, et al., 2022). Figure 3 depicts the influence of climate change on the interaction patterns of crops with ecological factors, which leads to crop damage and decreases in crop yield.

In the context of climate change adaptation, different tillage practices significantly impact soil organic carbon and nitrogen pools in rice-wheat cropping systems (Kumari & Singh, 2016). Conservation tillage and no-till practices enhance soil organic matter retention and improve nitrogen cycling, contributing to increased soil fertility and resilience against climate-induced stresses (Jagadesh et al., 2024; Kumari & Singh, 2016). These practices are essential for sustainable

agricultural productivity and effective climate change mitigation strategies in rice-wheat systems.

3 | INFLUENCE OF CLIMATE CHANGE ON GENETIC RESOURCES

The variety and variability of flora, fauna, and microorganisms used by farmers, forest dwellers, and fishers to produce food and nonfood agricultural products as well as to sustain ecosystem structures, functions, and processes within and around production systems are considered genetic resources for food and sustainable agriculture (Asfaw & Lipper, 2011; FAO, 2015; Gray & Brady, 2016; Hussain et al., 2020). Climate change is one of the major issues associated with the depletion of genetic resources, the foundational components on which local people and academics rely to increase the quality and quantity of food production (DaMatta et al., 2010; Datta et al., 2022). Newly developed varieties, breeds, and populations adapted to maximum production temperatures and enhanced or reduced amounts of rainfall are already being developed worldwide. Increased tolerance to abiotic stresses such as low and high temperatures, cold, limited water supply, soil flooding, and UV radiation is needed (Agnihotri & Seth, 2019). However, climate change also threatens the strategic stockpile of genetic resources for crops and livestock that will breed the new types necessary to adjust production systems to address upcoming difficulties (Altieri & Nicholls, 2017; Deryng et al., 2016; Abbass et al., 2022; Kumari & Singh, 2022).

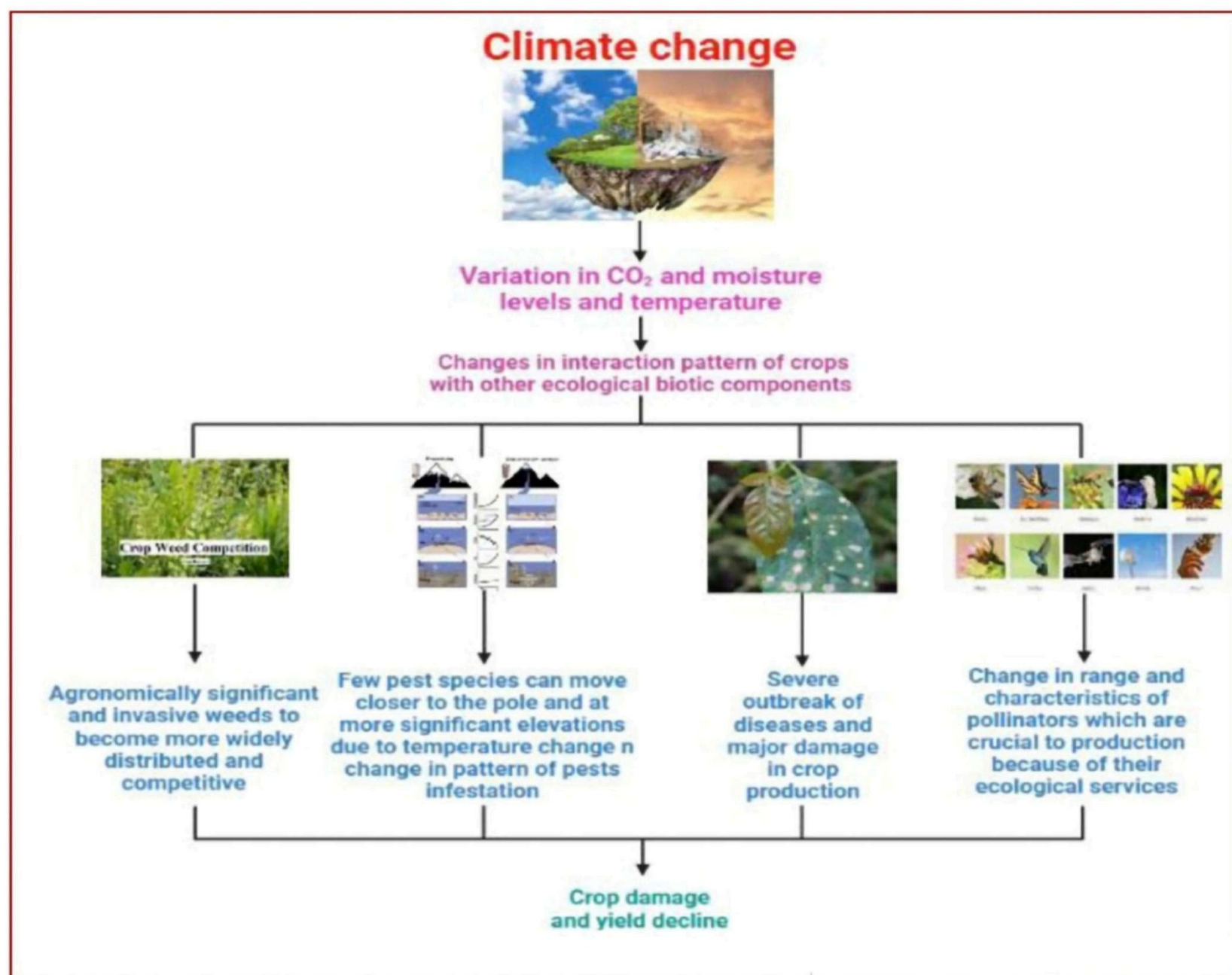


FIGURE 3 Impact of climate change on the interaction pattern of crops with ecological biotic components. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pce.15078)]

Farmers around the world are developing new varieties, breeds, and populations of crops and livestock that can thrive at extreme temperatures and with varying amounts of rainfall. These adaptations are necessary as climate change continues to pose a threat to the genetic resources that are essential for the future of sustainable agriculture (Abbass et al., 2022; Lusk & Polzin, 2023). Floods, droughts, and other extreme weather events are expected to occur more frequently and could lead to the permanent loss of certain varieties of crops and livestock. Researchers and breeders are developing novel breeding approaches and more effective use of genetic resources to increase the resilience of crops and livestock to climate change. According to plant biologists, various tree populations are able to withstand the effects of climate change (Datta et al., 2022; Kumari et al., 2018; Platt et al., 2020; Singh et al., 2013).

Wheat, which is one of the most important crops for human nutrition, is also being developed to withstand the effects of climate change. Researchers and breeders are using genetic sequencing and marker-assisted selection to identify disease- and fungal resistance

genes, as well as traits that can help crops cope with drought and cold temperatures. However, there are still many challenges that need to be addressed, such as how to keep pace with the rate of climate change, how to transfer knowledge and technology to farmers, and how to ensure a sufficient approach in research and development. Addressing these issues ensures that agriculture remains resilient in the face of climate change (Horcicka, 2015; Kumari et al., 2022; Lu et al., 2015; Malhi et al., 2021; Verma, Song, Degu, et al., 2023).

4 | ECONOMIC AND SOCIAL IMPLICATIONS

Most emerging and developing countries are at risk from climate change but also have less resources to adapt to its effects. Climate change will negatively impact agriculture, forestry, energy, and tourism and will depend on specific temperatures and precipitation levels (Altieri & Nicholls, 2017; Karavolias et al., 2021; Tripathi &

Mishra, 2017). The impacts of production are closely related to the economic effects of various proportions of the farm and the food chain, as well as to social ramifications. Climatic change has different social and economic implications and may modify agricultural revenues, food markets, prices, trade, and investment patterns. It directly affects farming households on a social level, which reduces their ability to pay for other expenses, such as health and education (Brown & Berry, 2022; FAO, 2023). At the national level, this approach can potentially increase the cost of agricultural products (food and feed), which affects the economic and social standing of the entire population, particularly in nations where food costs make up a significant portion of household budgets. It has macroeconomic repercussions for countries heavily dependent on agriculture, where agriculture contributes significantly to GDP (Khan et al., 2023).

These effects are anticipated to develop over time and with the advancement of climate change impacts and across various geographic regions and demographic groups. The implications of the current rise in the intensity and frequency of extreme events and weather anomalies, as well as the short- and long-term effects of significant changes in global warming, are described (Seth, 2012; Becklin et al., 2016; Habib-ur-Rahman et al., 2022; Ray et al., 2019). Since there is still only a small amount of related research in this field, and because of the limitations of international climate modelling studies, estimating the economic effects of long-term global warming is challenging. Another problem is uncertainty about the measures people can take to lessen unfavourably and increase positive outcomes. Weather patterns influence food production and reduce crop yield, migration, displacement, and food and water scarcity (Martinez et al., 2018). Climate change will adversely affect economic growth, and social conflicts are increasing overstressed ecosystems, leading to poor social and mental health. Suppose no solution is explored for mitigation and adaptation. This will pose serious risks to human health, global food security and sustainable economic development (Tripathi & Mishra, 2017).

5 | LIMITATIONS ON FOOD SECURITY AND SAFETY

Agroecosystems are directly impacted by climate change, which could influence agricultural productivity, affect the economy and society, and affect livelihoods and food safety (Altieri & Nicholls, 2017). In other words, the climate effect extends to the environment, the production sphere, the economy, and the social sphere (Raza et al., 2019). Depending on the weak areas at each stage of the system, the transfer of stress can be accelerated or inhibited. The IPCC reported (IPCC, 2014, 2023) that the broad range of social and economic processes and factors that have been poorly considered up to this point affects exposure and susceptibility, making it challenging to perform quantitative assessments. Additionally, it mentions how the dangers associated with climate amplify other stressors, which frequently affect people's livelihoods. The study of

climate change has focused on demographic, social, economic, and other traits related to risk exposure and resilience to adverse climatic conditions (Abbas et al., 2022; Jupinder et al., 2014; Lu et al., 2015).

6 | FOOD SAFETY ISSUES DURING CLIMATE CHANGE

The food systems required for food safety are susceptible to various threats. These dangers directly influence all facets of food safety and nutrition, including agricultural output, availability, use, and stability. In addition to the previously mentioned wide variety of other risks, which are either directly or indirectly affected by climate change or may interact with risks by climate change, compensatory, cumulative, or amplifying effects may occur (Deryng et al., 2016). The possibility that a food system would be unable to provide food security due to climate change is meant by the concept of "food security vulnerability" to climate change (Abbass et al., 2022; Brown & Berry, 2022; FAO, 2023; IPCC, 2012).

In 2014, the IPCC report mentioned the importance of governance in addressing vulnerabilities, citing examples of specific institutional vulnerabilities. Farms that produce only one crop are more susceptible to crop-damaging pests or a drop in prices, whereas systems with great diversity are less vulnerable to such issues. Regions experiencing frequent water shortages are more affected by drought, and in such areas, rainfed systems are more susceptible to drought than irrigated systems are (FAO, 2023; Khan et al., 2023). Families that rely solely on rain-fed agriculture for income are more economically vulnerable to drought than are those with alternative sources of income, particularly if they have no assets or social safety institutions. The effects of drought are transmitted from the biophysical to agricultural household levels (Datta et al., 2022; Karavolias et al., 2021). Rural residents constitute the majority of the world's poor, and the food-insecure population is directly or indirectly dependent on agricultural productivity for their livelihoods. Farmers and smallholders rely on off-farm labour and reduce food and nonfood expenditures to cope with these vulnerabilities (DaMatta et al., 2010; Sharma & Mujumdar, 2017; Skoufias & Quisumbing, 2005).

7 | RISK ASSESSMENT AND TOOLS TO ASSESS IMPACT OF CLIMATE CHANGE

Agriculture has a complex structure and interactions with different components, which will make future climate uncertainty a serious risk to food security and safety. Consequently, it is essential to assess the harmful impacts of climate change on agricultural crop production and develop adaptive applications to combat climate change (Datta et al., 2022). The era of climate change risk to food security and safety refers to the breakdown of food systems, that is, crop plant, livestock, and fisheries, as well as disruptions in the distribution of food, associated to global warming, severe water stress, soil flooding, and precipitation variability and extremes, particularly for populations already vulnerable to food insecurity due to limited income, lower

physical access to nutritious food, social discrimination or other factors (Mirzabaev et al., 2023). Simulation models, that is, general circulation models (GCMs) and representative concentration pathways (RCPs), are being used globally for the quantification of the negative effects of climate change on agriculture and are supporting the generation of future agroclimatic data (Arbuckle et al., 2015; Rahman et al., 2018). Crop models can potentially describe the internal processes of crops by considering the temperature increase and elevated CO₂ during critical plant growth stages (Challinor et al., 2018). There are no advanced approaches or strategies available to determine the impact of climate variability and change on the production of livestock and crops other than modelling (Asseng et al., 2014). Modelling tools are available and are being used worldwide to quantify the impacts of climate change and variability on crop and livestock production (Abbass et al., 2022; Ewert et al., 2015; Hoogenboom et al., 2015; Rahman et al., 2019; Verma, Song, Budeguer, et al., 2020).

To mitigate and adapt to the drastic effects of climate variability and change, information and communication technologies (ICTs) can also play a major role by promoting green technologies and less energy-consuming technologies (Bajwa et al., 2020; FAO, 2012; Kumari et al., 2023; Nizam et al., 2020; Shafiq et al., 2014; Zanamwe & Okunoye, 2013). Multiple barriers to adaptation exist that may limit the implementation and effectiveness of adaptation options. Effective adaptation is more than a series of discrete options (Mirzabaev et al., 2023). Timely provision of information from early warning systems (EWSs) and automatic weather stations (AWSs) on water deficit, floods, seasonal variability, and changing rainfall patterns can provide early warning about natural disasters and preventive measures (Imam et al., 2017), and it can also support farmers' efforts to minimize harmful effects on ecosystems. Geographical information systems (GIS), wireless sensor networks (WSNs), mobile technology (MT), web-based applications, satellite technology and UAVs can be used to mitigate and adapt to the adverse effects of climate change (Brown & Berry, 2022; Kalas, 2009; Karanasios, 2011; Lusk & Polzin, 2023). The application of different climate, crop, and economic models may also help reduce the adverse effects of climate variability and change on crop production [Table 1] (Abbass et al., 2022; Ewert et al., 2015; FAO, 2023; Habib-ur-Rahman et al., 2022; Hoogenboom et al., 2015). There are numerous specific tools and software applications that cater to different aspects of climate change adaptation, ranging from data analysis and modelling to decision support and communication. The choice of tools depends on the specific needs and objectives of a project or assessment in the context of sustainable agriculture (Table 1).

8 | CONCLUSION AND FUTURE PERSPECTIVES

Climate change poses a significant threat to agricultural sustainability, impacting crop productivity, food security, and socioeconomic stability. This review highlights the adverse effects of climate change on various aspects of agriculture, including temperature fluctuations,

erratic precipitation patterns, and increased pest and disease incidence. These changes necessitate the development and implementation of innovative adaptation and mitigation strategies to ensure agricultural resilience. Integrating climate-smart agricultural practices, such as the use of improved crop varieties, advanced breeding techniques, and sustainable resource management, is crucial. Leveraging technology and data-driven approaches, such as crop and climate modelling, can provide valuable insights for developing effective adaptation strategies. Collaborative efforts among researchers, policymakers, and farmers are essential to address these challenges and enhance agricultural resilience.

Looking ahead, several key areas warrant further research and development to enhance agricultural resilience to climate change. Continued efforts are needed to develop crop varieties and livestock breeds that are more resilient to climatic stresses. Promoting sustainable agricultural practices and integrating information and communication technologies, remote sensing, and precision agriculture can enhance climate monitoring and decision-making processes. Strengthening policies and institutional frameworks that support climate adaptation and resilience is crucial. Encouraging interdisciplinary research and international collaboration can facilitate knowledge sharing and best practices. By focusing on these areas, it is possible to build a more resilient and sustainable agricultural sector capable of withstanding the challenges posed by climate change.

ACKNOWLEDGEMENTS

The authors would like to thank the Guangxi Academy of Agricultural Sciences, Nanning, Guangxi, China, for providing the necessary facilities for this study. This research was financially supported by the Guangxi Natural Science Foundation (2021GXNSFAA220022), Guangxi Innovation Teams of Modern Agriculture Technology (nycytxgxcxt-2021-03), Guangxi Characteristic Crop Experimental Station (GTS2022022), National Key Research and Development Project (2022YFD2301102-07), The National Natural Science Foundation of China (31760415), Fund of Guangxi Academy of Agricultural Sciences (2021YT011) and Science and Technology Major Project of Guangxi (Guike AA22117002-1).

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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How to cite this article: Verma, K. K., Song, X.-P., Kumari, A., Jagadesh, M., Singh, S. K., Bhatt, R. et al. (2025) Climate change adaptation: Challenges for agricultural sustainability.

Plant, Cell & Environment, 48, 2522–2533.

<https://doi.org/10.1111/pce.15078>